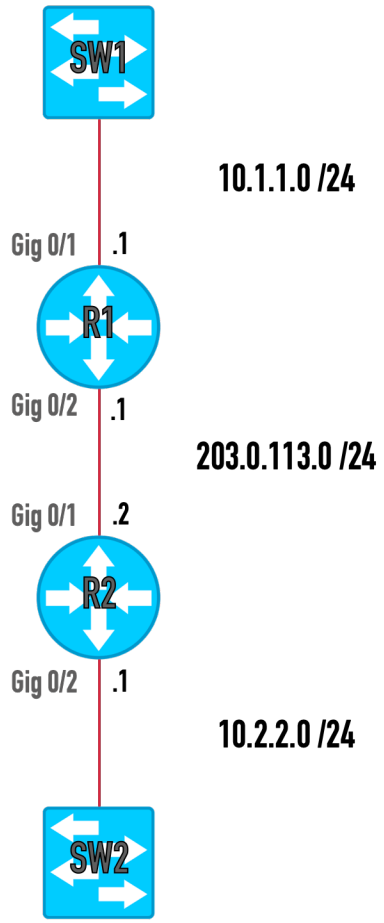
Embedded Event Manager (EEM)

Topology



Initial Configuration Commands

R1:

```
enable
conf t
host R1
no banner motd
no banner login
no banner exec
no banner incoming
line vty 0 15
password cisco
login
```

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```
line con 0
logging synchronous
exit
no ip domain-lookup
int gig 0/1
ip address 10.1.1.1 255.255.255.0
no shut
int gig 0/2
ip address 203.0.113.1 255.255.255.0
no shut
router ospf 1
network 0.0.0.0 255.255.255.255 area 0
end
copy run star
```

R2:

```
enable
conf t
host R2
no banner motd
no banner login
no banner exec
no banner incoming
line vty 0 15
password cisco
login
line con 0
logging synchronous
exit
no ip-domain lookup
int gig 0/1
ip address 203.0.113.2 255.255.255.0
no shut
int gig 0/2
ip address 10.2.2.1 255.255.255.0
no shut
router ospf 1
network 0.0.0.0 255.255.255.255 area 0
end
copy run star
```

(##This lab uses the same topology and base configuration as the IP SLA-Response Time Reporter (RTR) Lab.)

Lab Tasks

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1. Let's begin by enabling the "logging console" command on R1 to be able to view generated syslog messages.
2. Let’s go ahead and shut down the Gig 0/1 interface and then bring it immediately back up to see what syslog messages are generated. Go ahead and copy the syslog message that states “Interface GigabitEthernet0/1, changed state to administratively down”. We will use this later in our configuration.
3. Let’s go ahead and create an event detector. Choose “applet” and name it AUTO-NO-SHUT to enter applet configuration mode.
4. Let’s define an event detector. Since our focus is on detecting syslog messages, we’ll use the keyword “syslog” to identify an event. Then, we’ll utilize the keyword “pattern” to search for a specific string that will trigger an action. Inside the quotations, please paste the syslog message we copied earlier.
5. Now that we’ve set up the event detector, we’ll configure the actions to be taken via CLI commands when a trigger occurs. The first action is “enable”, followed by “conf t”, then “int gig 0/1”, and finally “no shut”.
6. Let’s test this by shutting down interface gig 0/1, and let’s see if it comes back up based on the actions that we configured. Let’s verify that that interface is still up by using the “show ip int brief” command.
7. Next, let’s create a new event detector, select “applet”, and name it NO-SHOW-RUN. This will take us back into applet configuration mode. This time, we’ll match a pattern based on a CLI input, specifically looking for any variation of “show run.” We want to “skip” this command and we don’t want this policy to execute synchronously.
8. Next, we are going to define an action and present users with a syslog message if they try to use the “show run” command. This syslog message will state, “Command not allowed, please contact an administrator.”
9. Let’s go ahead and test this out by issuing a “show run” command. Let’s see if our syslog message appears.

Solution

Step 1: Let's begin by enabling the "logging console" command on R1 to be able to view generated syslog messages.

```
R1>en
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#logging console
R1(config)#
*Sep 29 10:28:49.159: %SYS-5-LOG_CONFIG_CHANGE: Console logging: level
debugging, xml disabled, filtering disabled
```

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Step 2: Let's go ahead and shut down the Gig 0/1 interface and then bring it immediately back up to see what syslog messages are generated. Go ahead and copy the syslog message that states "Interface GigabitEthernet0/1, changed state to administratively down". We will use this later in our configuration.

```
R1(config)#int gig 0/1
R1(config-if)#shut
R1(config-if)#
*Sep 29 10:29:02.914: %LINK-5-CHANGED: Interface GigabitEthernet0/1, changed
state to administratively down
*Sep 29 10:29:03.916: %LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet0/1, changed state to down
R1(config-if)#no shut
R1(config-if)#
*Sep 29 10:29:10.466: %LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed
state to up
*Sep 29 10:29:11.468: %LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet0/1, changed state to up
R1(config-if)#exit
```

Step 3: Let's go ahead and create an event. Choose "applet" and name it AUTO-NO-SHUT to enter applet configuration mode.

```
R1(config)#event manager ?
  applet       Register an Event Manager applet
  detector     Set Embedded Event Manager detector information
  directory    Set Embedded Event Manager directory information
  environment  Set an Embedded Event Manager global environment variable
  history      Set Embedded Event Manager history information
  policy       Register an Embedded Event Manager policy
  scheduler    Set Event Manager scheduler options
  session      Set Embedded Event Manager session attributes

R1(config)#event manager applet AUTO-NO-SHUT
```

Step 4: Let's define an event detector. Since our focus is on detecting syslog messages, we'll use the keyword "syslog" to identify an event. Then, we'll utilize the keyword "pattern" to search for a specific string that will trigger an action. Inside the quotations, please paste the syslog message we copied earlier.

```
R1(config-applet)#event ?
  application         Application specific event
  cli                 CLI event
  config              Configuration policy event
  counter             Counter event
  env                 Environmental event
  identity            Identity event
  interface           Interface event
```

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```
  ioswdsysmon          IOS WDSysMon event
  ipsla                IPSLA Event
  neighbor-discovery   Neighbor Discovery event
  nf                   NF Event
  none                 Manually run policy event
  oir                  OIR event
  resource             Resource event
  rf                   Redundancy Facility event
  routing              Routing event
  rpc                  Remote Procedure Call event
  snmp                 SNMP event
  snmp-notification    SNMP Notification Event
  snmp-object          SNMP object event
  syslog               Syslog event
  tag                  event tag identifier
  timer                Timer event
  track                Tracking object event

R1(config-applet)#event syslog ?
  facility                Match string for facility
  maxrun                  Maximum runtime of applet
  mnemonic                Match string for mnemonic
  occurs                  Number of occurrences before raising event
  pattern                 Pattern match string for the entire message
  period                  Occurrence period
  priority                Screen messages that have specified priority
  ratelimit               Limit trigger to once per given time period
  sequence                Match string for sequence number
  severity-critical       Critical conditions, immediate attention needed
  severity-debugging      Debugging messages
  severity-fatal          System is unusable
  severity-major          Major conditions
  severity-minor          Minor conditions
  severity-normal         Normal event, signifying returning to normal state
  severity-notification   Basic notification, informational messages
  severity-warning        Warning conditions
  timestamp               Match string for timestamp

R1(config-applet)#$hernet0/1, changed state to administratively down"
```

(##Whenever we paste the syslog message we copied earlier, the CLI alters the command from "event syslog pattern 'Interface GigabitEthernet0/1, changed state to administratively down'" to the version you see here.)

Step 5: Now that we've set up the event detector, we'll configure the actions to be taken via CLI commands when a trigger occurs. The first action is "enable", followed by "conf t", then "int gig 0/1", and finally "no shut".

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```
R1(config-applet)#action 1.0 ?
  add                 Add
  append              Append to a variable
  break               Break out of a conditional loop
  cli                 Execute a CLI command
  cns-event           Send a CNS event
  comment             add comment
  context             Save or retrieve context information
  continue            Continue to next loop iteration
  counter             Modify a counter value
  decrement           Decrement a variable
  divide              Divide
  else                else conditional
  elseif              elseif conditional
  end                 end conditional block
  exit                Exit from applet run
  file                file operations
  force-switchover    Force a software switchover
  foreach             foreach loop
  gets                get line of input from active tty
  handle-error        On error action
  help                Read/Set parser help buffer
  if                  if conditional
  increment           Increment a variable
  info                Obtain system specific information
  mail                Send an e-mail
  multiply            Multiply
  policy              Run a pre-registered policy
  publish-event       Publish an application specific event
  puts                print data to active tty
  regexp              regular expression match
  reload              Reload system
  set                 Set a variable
  snmp-object-value   Specify value for the SNMP get request
  snmp-trap           Send an SNMP trap
  string              string commands
  subtract            Subtract
  syslog              Log a syslog message
  track               Read/Set a tracking object
  wait                Wait for a specified amount of time
```

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```
  while                 while loop

R1(config-applet)#action 1.0 cli command "enable"
R1(config-applet)#action 1.1 cli command "conf t"
R1(config-applet)#action 1.2 cli command "int gig 0/1"
R1(config-applet)#action 1.3 cli command "no shut"
R1(config-applet)#exit
```

(##It is a very standard convention to name these actions in a numerical sense. When we have more actions, we will number those sequentially.)

Step 6: Let's test this by shutting down interface gig 0/1, and let's see if it comes back up based on the actions that we configured. Let's verify that that interface is still up by using the "show ip int brief" command.

```
R1(config)#int gig 0/1
R1(config-if)#shut
R1(config-if)#
*Sep 29 10:36:33.994: %LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to
administratively down
*Sep 29 10:36:34.996: %LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet0/1, changed state to down
R1(config-if)#
*Sep 29 10:36:36.308: %LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to
up
*Sep 29 10:36:37.310: %LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet0/1, changed state to up
R1(config-if)#end
R1#show ip int brief
Interface                  IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0         unassigned      YES TFTP   administratively down down
GigabitEthernet0/1         10.1.1.1        YES TFTP   up                    up
GigabitEthernet0/2         203.0.113.1     YES TFTP   up                    up
GigabitEthernet0/3         unassigned      YES TFTP   administratively down down
```

(##As you can see, when we attempted to shut down interface Gig 0/1, it came back up due to the policy and actions we configured.)

Step 7: Next, let's create a new event detector, select "applet", and name it NO-SHOW-RUN. This will take us back into applet configuration mode. This time, we'll match a pattern based on

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a CLI input, specifically looking for any variation of “show run.” We want to “skip” this command and we don’t want this policy to execute synchronously.

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#event manager applet NO-SHOW-RUN
R1(config-applet)#event cli pattern "show run" ?
  default       default action timer
  enter         Event match upon enter key
  maxrun        Maximum runtime of applet
  mode          CLI parser mode
  occurs        The number of occurrences before raising the event
  period        Number of occurrences must occur within this time period
  questionmark  Event match upon questionmark key
  ratelimit     Limit trigger to once per given time period
  skip          Whether to skip CLI command execution
  sync          CLI and EEM policy execution sync or async
  tab           Event match upon tab key

R1(config-applet)#event cli pattern "show run" skip ?
  no   CLI command should be executed
  yes  CLI command should not be executed

R1(config-applet)#event cli pattern "show run" skip yes ?
  default    default action timer
  maxrun     Maximum runtime of applet
  occurs     The number of occurrences before raising the event
  period     Number of occurrences must occur within this time period
  ratelimit  Limit trigger to once per given time period
  sync       CLI and EEM policy execution sync or async

R1(config-applet)#event cli pattern "show run" skip yes sync ?
  no   Policy and CLI will run asynchronously
  yes  Run policy and the result determines whether to run CLI

R1(config-applet)#event cli pattern "show run" skip yes sync no
```

(##Running this policy asynchronously means it will execute before the “show run” command, reducing the chances of the “show run” command generating any output.)

Step 8: Next, we are going to define an action and present users with a syslog message if they try to use the “show run” command. This syslog message will state, “Command not allowed, please contact an administrator.”

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```
R1(config-applet)#action 1.0 ?
  add                 Add
  append              Append to a variable
  break               Break out of a conditional loop
  cli                 Execute a CLI command
  cns-event           Send a CNS event
  comment             add comment
  context             Save or retrieve context information
  continue            Continue to next loop iteration
  counter             Modify a counter value
  decrement           Decrement a variable
  divide              Divide
  else                else conditional
  elseif              elseif conditional
  end                 end conditional block
  exit                Exit from applet run
  file                file operations
  force-switchover    Force a software switchover
  foreach             foreach loop
  gets                get line of input from active tty
  handle-error        On error action
  help                Read/Set parser help buffer
  if                  if conditional
  increment           Increment a variable
  info                Obtain system specific information
  mail                Send an e-mail
  multiply            Multiply
  policy              Run a pre-registered policy
  publish-event       Publish an application specific event
  puts                print data to active tty
  regexp              regular expression match
  reload              Reload system
  set                 Set a variable
  snmp-object-value   Specify value for the SNMP get request
  snmp-trap           Send an SNMP trap
  string              string commands
  subtract            Subtract
  syslog              Log a syslog message
  track               Read/Set a tracking object
  wait                Wait for a specified amount of time
  while               while loop

R1(config-applet)#action 1.0 syslog msg ?
  WORD  message string

R1(config-applet)#$ "Command not allowed, please contact an administrator."
R1(config-applet)#end
```

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(##The original command we entered before the CLI changed it to what you see here is: "action 1.0 syslog msg 'Command not allowed, please contact an administrator.'".)

Step 9: Let's go ahead and test this out by issuing a "show run" command. Let's see if our syslog message appears.

```
R1#show run
R1#
*Sep 29 10:41:56.711: %HA_EM-6-LOG: NO-SHOW-RUN: Command not allowed,
please contact an administrator.
```

(##You can see here that the command is not allowed because of the policy that we put in place for the "show run" command.)

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